

Transport and sustainable democracy: assent and system dynamics

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Familiar paradoxes, elusive solutions

The electoral history of our democracies shows that it is difficult to achieve a genuine reconciliation of individual preferences and collective choices. Voting systems, the objective limitations of which have been familiar since Condorcet and Arrow, do not always produce choices that reflect the opinion of the majority.

In response to these familiar concerns, a range of experiments have been conducted with alternative voting systems. For instance, a voting system based on assent has recently been tested in France (Balinski 2002). Its basic principle – to give the voter the opportunity to support several candidates instead of just one – is designed to promote an alternative conception of democracy and public choices.

What underlies this approach is the idea that preferences are not in fact completely exclusive: each individual can feel “represented” by a range of possible combinations of choices. Indeed, the system recommended and tested is voting “by assent”. Tests conducted during the French presidential elections indicate that the result of such a procedure differs from that of standard competitive voting. Specifically, it modifies the relative weight of the candidates

and, in some cases, their ranking. Furthermore, assent gives greater prominence to certain unusual candidates who suffer from traditional forms of “tactical” voting. Thus, the balance of votes may be modified so as better to express ideas, rather than merely protest. The value of such a method of consultation is particularly clear in the case of transport, where apparent conflicts between available options are sharp and protest is both common and generally discredited.

Similarly, urban issues give rise to sharp political cleavages, which represent contrasted overall logics of development that can be summarised by alternative scenarios. It makes sense, therefore, to think of local democratic debate as being organised around competing high-level conceptions of what is at stake in “mobility”.

With this in mind, Balinski (2002) sketches four “caricatures”, which he proposes to go beyond by calling upon a fifth conception or “scenario”, that of “*homo civis*”, the usefulness of which he summarises as follows: “scenario 5 is less of a caricature than the others. It is explicitly developed with a view to realism and accessibility. Unlike the first four, which are reductionist by design, it aims at convincing”. He adds, a little later in the text, “It is thus the scenario of wisdom”. Nonetheless, the

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organisation of debate and political take-up of hypothetical scenarios raise complex issues: urban policy problems are “transactions” that rely heavily on “mediations”. “The logic is collective because individuals are not made directly responsible for their choices.” The connection between individual and collective thus requires mediations, which “are based on expression of, and confrontation and compromise between, competing needs, not directly by individuals but by the bodies that represent them”.

It is obvious enough that the various components of a town or a conurbation have neither the same immediate interests nor the same priorities. Hence the idea of a “transaction”. This, however, immediately raises a three-fold question of legitimacy, along with an analytical requirement:

- the legitimacy of institutions and territorial divisions, which are the basis for part of the transactions and compromises,
- the legitimacy of the distribution of competences,
- the legitimacy of the “representative entities” that are superimposed upon institutional representative processes,
- analytical clarification of interactions between the actions or decisions of the relevant collectivities.

With respect to urban transport, local competence is fairly easy to justify, and the legitimacy of local authorities is little challenged – although centralised systems do continue, in some cases, to burden the actual organisation of passenger transport. Outside the urban context, on the other hand, as with respect to freight generally, things are more complex. Furthermore, there are obviously strong interactions between the organisation of local passenger transport and freight transport.

For example, congestion and competition for road space and railway tracks are manifest issues for local policy, and may have major impacts on the efficiency of freight transport.

To possible conflicts between local interests, or between community goals at different levels, should be added divergences of opinion or analysis. From this point of view, “subsidiarity” cannot simply be “taken for granted” (using the

phrase in the usage familiar from ethnomethodology): it does not offer a basis for the political cohesion of the “group”. Different levels of multi-rationality may lead to total indeterminacy, or at least to decisions that do not address the problem at hand, but rather arbitrate partial clashes of interests.

Within the European Union (EU), this complex reality, made up of multiple inter-relations, currently comes up against two apparently opposite dynamics: European integration, on the one hand, and the emergence of local and identity politics, on the other.

Two conclusions follow.

- Firstly, consultative or decision procedures based on assent rather than exclusive choice may facilitate the emergence of collectively preferable outcomes.
- Secondly, “honest” transactions between relevant entities require a dynamic apprehension of the multifarious inter-relations between actors and their choices.

It still remains, however, to get a clearer grasp of the real impact of the institutional and political development of Europe.

Community development and local and identity claims

The dual process of European Community development and the growth of local and identity claims, demands, and mobilisations, has a major impact on the conditions of collective choice about transport issues.

On the one hand, European integration leads to the growth of common rules and practices. The major areas of transport are all now within the single market, which reinforces the legitimacy of a single policy and, as a consequence, a single authority. From being technical, regulatory, economic, fiscal and social, integration is becoming societal and cultural. Europe is now a reality for transport by sea and by air, for road transport as for river and rail. What is currently at stake is the need for a Community authority, in other words an executive body with direct regulatory capacity, alongside growing harmonisation of national legislation. Understandably, this issue arises

first in response to safety concerns (air traffic, inspection and control of ships and coast, etc.), but subsequently focuses on methods and means to implement major infrastructure options in a coherent way, as well as to defend the interests of the Community as a whole vis-à-vis the rest of the world (in areas such as air transport negotiations, management of major international rivers, etc.).

At the same time, local or identity-based demands have acquired new significance. One reason is, undoubtedly, growing sensitivity to environmental problems, as well as to the issues raised by urban development, which naturally point to the legitimacy of a local political arena. Furthermore, three-quarters of all EU legislation is implemented at local or regional level. It is therefore logical enough that, in the words of the EU Committee of the Regions (which was established by the Treaty of Maastricht in 1992), “representatives of local and regional authorities should have a place in the design of new Community laws”. There is, however, another dimension: in very general terms, there is currently a presumption in favour of the legitimacy of strong local policies, and conversely suspicion about excessively global rules, whether national or European. The pressure for greater decentralisation is of course especially visible in a traditionally centralised country such as France. It is based on stronger subsidiarity and greater respect for regional identities, which can in turn present themselves, through autonomist or separatist mobilisation, as having the potential to become national.

These two powerful trends, driven by the speed of certain economic transformations, translate into a crisis of institutional organisation and to political protest in defence of national sovereignty, which is perceptible – albeit in a volatile way – in the electoral dynamics of the various EU member states. In this respect, the fear that national governments and representatives no longer “control” anything is doubtless added, in some countries such as France, to the effects of a constitution that gives predominant power to the executive. This is particularly significant in freight transport, where policy was mainly the responsibility of the state, and where debates about the organisation of transport or major infrastructure projects had

long ceased, for reasons unrelated to Europeanisation, to take place in Parliament.

The main statutes and instruments that provide the basis for sectoral regulation are now entirely dissolved in the unifying architecture of the Community. Powerful public sectors based on natural geographical monopolies or on strong ideas of public services are now being challenged, in their very nature, by the logic of a process of European integration based on the principle of non-discrimination. They are seen as obstacles both to the construction of Europe and, more seriously, to the implementation of common policies.

Although the implication of member states may, in the past, have been often more virtual than real, it is undoubtedly now in decline, while at the same time there are increasingly explicit calls for common policies and a common authority. In road transport, the key development in this respect was the Single European Act of 1986, which enshrined the principle of the abolition of internal borders, and thereby abolished any basis for bilateral regulation of road freight haulage.

Conversely, rail transport is currently at the symbolic core of European issues, not simply because it is in crisis at the very time when European policy requires it to expand, but because it remains largely trapped within national policy logics, and their corollary, which is bilateralism.

Other indications of the dual tendency at work can be found in the management of Alpine issues, as reflected in the “transport protocol” of the Alpine Convention, which was concluded in 2000.¹ What is at work here is a movement – an incoherent one in institutional terms – towards community competence (in other words, Europeanisation), even as the legitimacy of the process itself derives from territorial specificity. It is as if bottom-up subsidiarity required Community justification.

The political impact of these developments should not be underestimated. A feeling of complexity, combined with the dissolution of existing social frameworks and representations, is emerging without new forms of legitimacy emerging to take the place of the old in public perceptions.

Hence the importance, particularly as regards transport, of developing new ways of

consulting collectivities and social groups, while permitting a broader analysis of underlying realities that might enable all concerned to grasp the interactive nature of the system.

What place for transport in the Union – state – territory system ?

Given that transport is, in many ways, characteristic of contemporary political problems, two sets of issues deserve further attention.

General transport policy

Overall transport policy options – as represented, say, by Commission white papers – would seem to fall legitimately within the competence of the EU. As mentioned earlier, the transfer of competence over the past 20 years has been complete. The general spirit of policy, action principles, the regulatory framework and basic legal principles, as well as principles of organisation, are no longer within the purview of national policy. The true policy process, with respect to European transport, lies in the “co-decision” procedures of the Council of Ministers (which comprises representatives of member state governments) and the Parliament (which comprises representatives of national electorates). Implementation, on the other hand, is primarily within the competence of member states, which transpose, adapt and apply – more or less faithfully – what has been decided in common.

However, institutional trends are giving this set-up a changed and fluid meaning.

The phrase “in common” naturally refers to the decision mechanisms of the Council, which represents states. But co-decision, as introduced by the Treaty of Amsterdam in 1997, introduces two changes. First, “in common” now refers to the search for compromise between two sources of EU legitimacy – the Council and the Parliament – as well as pointing to the common legislative work of Parliament itself. The new decision procedures significantly shift the centre of gravity of the system toward the Union and away from the Community of states.

In the final analysis, what are at stake in such a context are the effectiveness and the overall coherence of national policies.

The example of rail policy is characteristic in this respect. The gap between Community objectives, directives, often belated transposition, and actually implemented national policies, reveals contradictions that eventually lead, via lengthy feedback loops, to amendment of European objectives and directives. This shows that formal acceptance of a set of objectives and legal instruments is not the same thing as implementation and, crucially, that real national powers remain considerable. Hence a widespread feeling of crisis and inefficiency. In this case, the management and implementation of common policies are perhaps not quite consistent with the initial organisation of the sector.

In fact, solutions to these difficulties depend on the good will of the member states. Political and trade union obstacles are simply too powerful to be ignored. As a result, it is as if, consistently with “partial conflict resolution” as formulated by organisational sociology, governments on the one hand agreed on principles at European level in order not to appear “anti-European”, while simultaneously, on the other hand, implementing them weakly or not at all, in order to avoid clashes with national opponents.

It then falls to political discourse to offer an *ex post* rationalisation of this irrational position. Thus, French policy language between 1997 and 2002 sought to reconcile European integration with the rail *status quo*. For example, Jean-Claude Gayssot, the then Transport Minister, stated in a magazine interview in March 1998: “we have established rail freight corridors that respond to a quite different logic than the Commission’s White Paper. It must be understood that, in doing so, we are facing up to a genuine challenge. We must now prove that our alternative works and that we are capable of developing European rail traffic”.

Such an approach leads to what Cyert & March (1963), following Herbert Simon, call “partial rationality”. In this case, the issue is to adopt two different postures, in Brussels on the one hand, in Paris, London, or Berlin on the other, responding to two distinct objectives. The problem is that the objectives can be reconciled only over time, by tougher European legislation

or enhanced supranational powers. However, while the feedback loop is lengthy, we may suppose that it is gradually shortened by the deepening of European policy, which limits states' room for manoeuvre, and by the sequential integration of increasingly divergent objectives. If this view is correct, ever deeper Union is not merely inevitable, but ever more inimical to partial conflict resolution strategies. The path, nonetheless, is a tortuous one.

This is one reason why the transport White Paper (European Commission 1996) makes the following key statement: "The objective of restoring balance to the transport system, which has hitherto proved unattainable, presupposes not merely the implementation within the common transport policy of the ambitious programme of measures proposed by the White Paper by 2010, but also consistent measures implemented by other policies at national or local level." In other words, states must play their part, along with local authorities. And it is at the local policy level that further questions, to which the population are highly sensitive, are raised.

Local realities

The task of transposing directives, reforming railway companies and changing tax regimes generally falls to states and involves a sometimes complex political, legislative, and regulatory labour. And this clearly raises some serious problems. What kind of political power is it the main function of which is to serve up policies defined "elsewhere" – elsewhere being in this case all the further that there may be a strategic incentive to stay at a distance from it or even to denounce it almost explicitly? In such a case, state power derives much of its legitimacy from its capacity for resistance.

With respect to the national aspects of this problem, the European Commission's verdict was, as early as 1996, definitive. "The management of rail systems has been the main factor in their decline. However, the difficulties encountered by rail transport have much to do with relations between states and railway companies. States have generally refused railway the freedom that commercial enterprises enjoy. As well as short-term political interference, the autho-

rities have tended to demand the maintenance of services considerably below the threshold of profitability. Rail investment has often been inadequate or poorly directed, and has consequently been a burden on public finances. Governments have compensated for losses by large subsidies with no specific objective, such as enhanced efficiency. Furthermore, financial objectives were often vague".

The summing up is harsh. It reflects a clear conflict of interest based on a fundamental incompatibility of objectives. It is on the basis of the failed implementation of the directives of 1991 that many of the recommendations of 1996 were derived and, later, that analysis and decisions were pushed in a more radical direction in 2001.

Relations with the local level, however, are more complex. Transport is organised at a macro-level, but implemented locally. Communication paths and port or airport zones interact with a physical space filled with tangible inhabitants. Pollution and congestion are localised phenomena. And, while general transport policy often remains regrettably within the parameters of sequential management of contradictory objectives, local issues are more commonly set within a framework of "displacement". It is obvious enough that a flow that cannot go through the Mont Blanc tunnel will necessarily find another path, that town-centre measures affect the suburbs, and that fiscally profitable land development in one area has consequences for its neighbours. The regulatory capacity of local authorities is, in the final analysis, a key mediator of major European policies.

From this follows the complex problem of relations between levels of competence and of their proliferation. It is hardly surprising that such a complex system as transport evades strategic coherence since, precisely, organisational complexity is an ad hoc method for decomposing incompatible objectives into a apparently manageable problems.

In France, the superposition of Europe, the state, the regions, the departments, the inter-communal authorities and the communes produces decisions that may well be "locally acceptable", but may nonetheless be, at the same time, inefficient or even absurd at the aggregate level. This leads inevitably to the



Off the ferry, Bastia, Corsica, August 1999. François Grabrielli/AFP

question how subsidiarity is to operate in an area where spatial interdependencies are strong, and, correlatively, to the appropriate methods of conflict resolution.

Road siting, traffic rules, port investments and nature conservation necessarily imply trade offs between community “levels”. Negotiations and political transactions necessarily take place between groups with both contradictory and complementary interests. Should the number of groups be defined or restricted? Should communities be institutionalised? Should arbitration rules and hierarchies be established? Should, on the contrary, contractual mechanisms be used exclusively? These are key questions, which bring into play the legitimacy of the various levels and of project evaluation procedures.

It is common enough that a change of parliamentary majority should lead to cancellation, or euphemistically re-assessment, of a

major project. Thus, in 1997, one of the first decisions of the incoming French government was to reverse the decision to build a canal linking the Rhone to the Rhine. Similarly, in 2002, the then opposition announced shortly before its election victory that “the planned third Paris airport [would] be re-assessed on an ‘objective and transparent’ basis”. This reversed the process relaunched by the then government, which had itself reversed the decision of its predecessor. On this infrastructure project, two decisions were thus overturned in five years.

Infrastructure issues are clearly the object of most visible public decisions, and therefore of the political debates that can challenge the government itself. Thus, the re-opening of the Mont Blanc tunnel to lorries was made a campaign issue by some candidates in the 2002 French presidential election, and was, a few weeks later, a major issue in the Alps during the parliamentary elections.

Taking full account of infrastructure debates comes up, as these examples show, against an institutional problem as well as a problem of analysis of options and project evaluation.

Powers, relations between powers, and the structuring of social demands

The problem here is a classic one of power and the structuring of social demands. An airport, for instance, is never built for those who live in the vicinity, but on the contrary for a broad community. Such is the nature of collective choices and large projects. However, where marginalist public choice theory offers fairly clear-cut answers, the political reality of transactions between social groups produces a sometimes inextricable tangle. It is clear that it is not enough to point to the advantages of the airport for the collectivity compared to the nuisance for local residents to justify public choice and underwrite the greater legitimacy of the “higher” community. Nor does incorporation of sustainable development concerns suffice to reduce all the contradictions.

Methods of analysis and consultation are of major significance here, but one should not neglect the broader question of political and institutional organisation, which raises particular problems in a traditionally “Jacobin” state such as France.

In other words, while actors’ information and socio-economic project evaluation are essential, the central questions are power and the legitimacy of social demands. This appears very clear with respect to the text-book case of Corsica, which has been a central and deeply divisive issue in French politics, especially since the mid-1990s. It is a particularly interesting case because it raises practical questions about legal authorities and their organisation with respect to transport, while also referring back to a broader debate about capacities and competences between central government and regional authorities. First, however, a word is necessary about the legislative framework and its recent innovations.

The Law of January 22 on Corsica enacts several important provisions with respect to transport.

- First, the Law institutes a “Planning and Sustainable Development Framework”, which is defined by the Corsican Executive Council and passed – after being referred to the consultative Corsican Economic and Social Council – by the Corsican Assembly. The Framework, which is subject to a public inquiry, expresses a “local” strategy, but must “take account of” projects or public interest or national concern (*d'intérêt national*). Once approved, it may lead to contractual relations with central government, and modifications may be requested by the representative of the state in order to permit a project of national concern. Provision is made for the Government to modify the Framework if its initial request is not complied with. The statute – which grants broader powers to a local authority than is usual in France – defines a fairly clear dividing line: to local authorities, planning, on condition it does not interfere with projects of national concern. Of course, such power sharing is meaningless unless accompanied by a corresponding decentralisation of competences.
- Such is precisely the purport of the Law, which empowers the Corsican authorities to impose public service obligations on sea and air transport operators serving the island, to create on its own initiative personal social assistance for the benefit of certain categories of passenger, and to delegate the provision of public services. Furthermore, the authority is fully competent over ports ‘with the exception of police powers’, creation and modification of airfields, and the rail system (including ownership of the network, which has been transferred to it).

This legislation illustrates the natural area of competence of a “local authority”: passenger transport (there is currently no rail freight in Corsica) and major infrastructures. By contrast, freight regulation and implementation and supervision and monitoring of national and Community regulations remains within the purview of the state (centrally, or through its regional offices). The Framework thus assumes that, with respect to freight, the only capacity the local authority has to define and implement a policy is, precisely, its control over infrastructure.

With this in mind, it is of interest to consider how elected members of the Corsican Assembly understand and interpret this form of power sharing, by looking at debate on the exceptional investment programme, which was implemented in response to the provisions of the Law (articles 53 and following). For analytical purposes, the intervention of one Assembly member deserves lengthy quotation.² He began as follows:

Where are the preliminary studies that allows us to disregard, for instance, the creation of railway line in the Eastern plain? Which technical, financial or economic studies ground this option at a time when in the whole of France, indeed in the whole of Europe, there is an emphasis on rebalancing transport between road and rail? Where are the studies? Where are the profitability assessment that make it possible to claim, in passing, that a tunnel will be built at Vizzanova, that it will be profitable, that it is worth building, whereas we know that, given current safety requirements, a two-tube tunnel will be inordinately expensive, to say nothing of operating costs, which are not even mentioned here (safety and monitoring teams 24 hours a day), all this for a level of current and potential traffic that is not even cited in the document, but hardly exceeds 2000 vehicles per day, which is far less than the Bastia South exit?

Here, the Assembly member neatly summarises the problem of the central state in a decentralised system and of the appropriation by new local authorities of their powers. He points out correctly that, in establishing an exceptional plan, the state cannot exempt itself from its usual procedures. He went on:

I would remind state officials, who seem to hope that such funding will be easy to obtain, since they claim there will be no difficulty in getting an inter-ministerial fund, that in this case it will be necessary to request money from the Infrastructure Ministry [*Ministère de l'Équipement*] and from the road department, and that, until further notice, national road investment requires economic evaluation (circular of October 20 1998), which isn't an option or an economist's fad; that, as a result, we shall not fund, I guarantee this today, a road project of which the economic bottom line, meaning the discounted surplus, would not be at least positive, to say nothing of negative. For that matter, even if it was funded, which I doubt, it will be very difficult to get a declaration of public interest, if the economic assessment and the overall assessment are not satisfactory. That is why I believe that today, before identifying major options (and indeed the convention says so), we should be careful to justify them by study and by calculation.

In passing, we may note that ambitious planning, involving significant transfers of competences, by no means absolves those responsible of rigorous procedures for analysis and evaluation. The Assembly member concluded that the priority is to provide resources:

not just financial, but also administrative, so that units can be established with the capacity to study, to prepare realistic options, to table them before elected representatives, and to launch public debate, which must also encompass civil society and the population.

Returning to the architecture and articulation of competences, it appears clearly that the reality of power depends on both the devolution of resources (which is a classic demand) and an indispensable capacity for expertise – if only with a view to adequate preparation of plans – and for leverage with respect to central government action and the content of contracts between the state and local authorities. In addition, one grasps here the lack of a tool to confront expertise, scenarios, and strategies.

What is difficult enough in the case of an island economy, which is set within natural geographical limits, becomes extraordinarily complex in “continental” collectivities, where there are major problems of localisation, siting and interactions with neighbouring regions. And this difficulty is no way peculiar to the national arena. Interlocking considerations have no reason to stop at the border. Nonetheless, it is striking that the French authorities did not discuss with their neighbours the “service frameworks” (*schémas de services*)³ that have now replaced the traditional infrastructure plans.

While designed to promote coherence, cooperation and common expression of policies, the EU is unable to create or to foster practices that might contribute both to better forms of globalisation and to better incorporation of local strategies. What works poorly, or not at all, between states does not necessarily work better – or worse for that matter – in traditionally centralised states groping towards devolution.

Evaluation at the heart of interfaces and trade-offs

If the problem derives in part from political and institutional cultures and practices, it is obvious

that the conduct of evaluation studies is crucial. This requires both awareness of the need for competence and the actual use and mastery of methods for adequate assessment of available options. In these terms, the “complex” that affects local politicians is perhaps not entirely justified. A recent survey of senior officials at the French Transport Ministry and Directors of public bodies within the sector highlighted their discomfort when faced with the major challenges of freight transport and the need to fulfil the state’s role, in relation to the profession as well as to local elected officials. There is in fact a general perception of a gap between expertise or technical capacity and social demands.

To make political compromises, to conduct transactions between social groups, to confront local interests and broader policy, to define and to implement medium-term strategies, and to clarify their environmental, economic and social stakes, are all issues that require evaluation studies. However, these prove difficult to execute precisely where they are most useful, or even indispensable. What was once simply a difficulty in overcoming purely sectoral or modal views, or corporate narrow-mindedness (not least in the public sector), has now become a general difficulty in dealing with multiple strategies so as to produce overall coherence.

Project evaluation, which has traditionally been tightly controlled by state corporations or agencies, is rarely conducted in a pluralistic way and rarely allows for cross-examination of pre-defined solutions. This situation is not viable in a largely decentralised EU system, on pain of profound frustrations. It is in this sense that European integration and its deepening, along with growing decentralisation, make necessary a major expansion of evaluation and analysis in a pluralistic context.

The systemic approach and strategic planning

Beyond the general requirement of pluralism – concerning expertise as well as opinions –, a resolutely systemic approach is unquestionably called for (Karsky & Salini 2000, Salini 2000a, 2000b, 2001).

Yet, curiously, there are few evaluation or overview studies that offer decision makers, and ultimately elected representatives, a global,

systematic, non-linear perspective on what “might happen” subject to various assumptions. In fact, there is a major obstacle to adequate analysis within traditional methods: the failure to take account of the numerous feedback loops that characterize the behaviour of even the smallest systems. This is what was earlier called the “interactivity” of decisions and policies. Let us take an example.

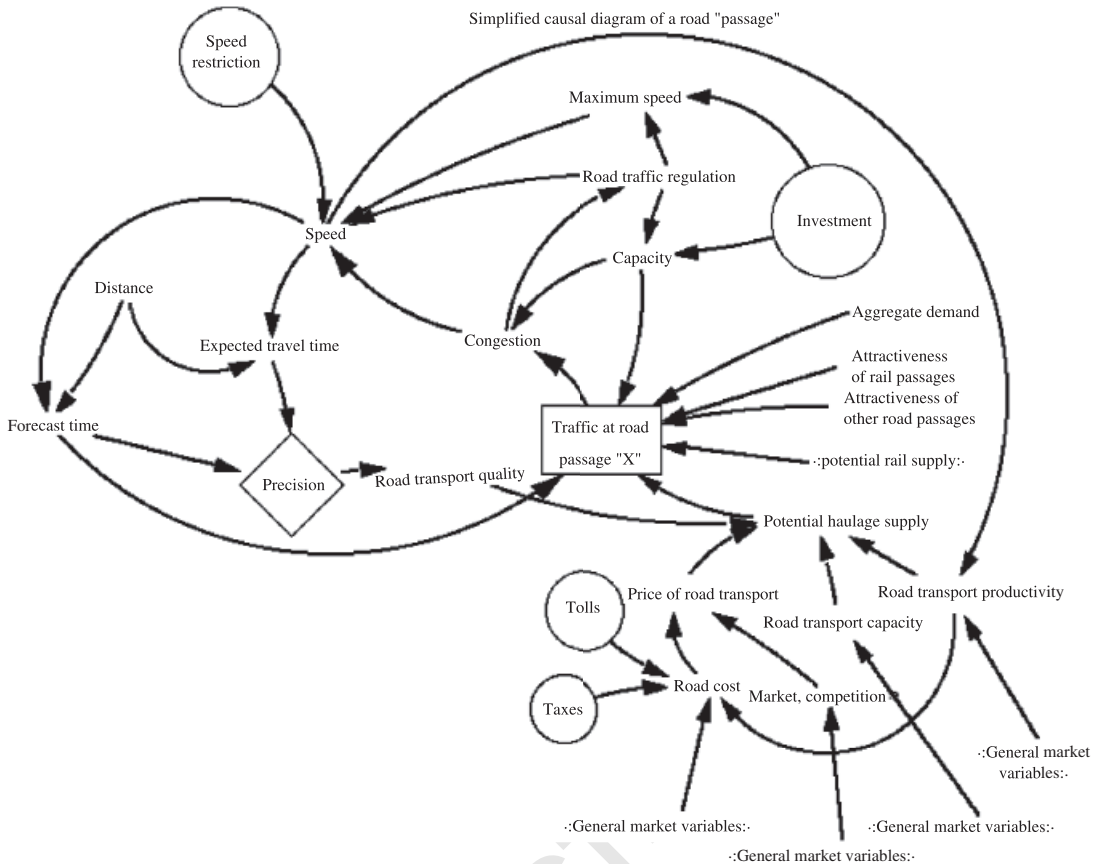
Considering Alpine policies, one would expect national and local politicians from the various countries to demand tools to evaluate their policies (whether modal, national, local, etc.). Institutionally, decision and consultation structure have no reason to be comparable or compatible from place to place. If, nonetheless, decision makers wish to organise consultation with their neighbours, if only to take account of their working assumptions and their likely strategies, what tools are available?

A cursory survey of what is “going in” in a tunnel or at a road passage, for example, gives some idea of the scale of the problems.

In practice, the alpine system can be “summarised” as a combination of some 20 subsystems similar to one represented in Fig. 1. Even this small system exhibits numerous elementary loops. Some are fairly obvious. The first, of course, involves the relations between excess traffic and congestion. In a familiar way, congestion increases with traffic flow, reducing average speed, and finally persuading motorists not to use the tunnel, which, ultimately, reduces traffic flow. This loop, which counteracts the effects of excess traffic, operates quite quickly so long as alternative routes are available.

Traffic regulation schemes have a direct impact on the average or commercial speed of heavy goods vehicles (HGVs), which ultimately affects the attractiveness of the tunnel as a route, which affects traffic flows, which in turn impact on regulation. In this way, regulatory intervention can increase average vehicle speed. Other things being equal, this should increase the attractiveness of the tunnel by reducing expected transit time. Figure 1 illustrates how this feedback loop, which is simple and fairly rapid, is located within the overall interaction system.

Other mechanisms will also operate, such as in-tunnel traffic regulation. Imagine that the tunnel manager triggers a regulatory scheme in response to congestion that permits an increase



Q1 FIGURE 1.

in average HGV speed. A loop (top right in Fig. 1) then operates to complicate a relationship previously analysed as direct (top left of Fig. 1). In this way, regulation, which may be implemented by an automatic system or by human decision, ultimately increases the attractiveness of the tunnel, and hence traffic flows. Regulation has thus reduced the negative impact on congestion on the attractiveness of the tunnel, but at the cost of increased congestion.

In fact, even considering only traffic regulation, it is clear that, at the same time, the managers of other tunnels or competing routes may themselves make decisions. Some might opt for fine tuning of regulation, others may prefer less frequent interventions. Even with respect to this one variable, there is thus a complex system involving several feedback loops, with different rhythms in different places. Only simulation

offers some predictive capacity taking account of the behavioural routines of the various actors.

Furthermore, not only do behavioural mechanisms involve numerous intermediate variables, there are also a wide range of action variables that each actor will seek to manipulate in response to specific objective. In Fig. 1, 16 loops pass through the variable "traffic flow at point X". No one has the ability to grasp the whole of this interaction and of its impact on traffic over time. No doubt some of these loops are simple enough, but even then their number and the fact that they operate simultaneously makes analysis complex. In addition, some are considerably more extended in time.

Imagine now that an increase in traffic flows leads to increased congestion. As a result, HGV speed is diminished along with, ultimately, vehicle productivity. The decrease pushes up

transporters' unit costs, and therefore their prices, which reduces their potential supply on the relevant transport axis, and therefore pushes traffic flows down. One should note that this loop is much longer in terms both of the number of variables it involves and the time it takes the feedback to operate.

What is interesting in these long interactions is that they may have major impacts even though their cause has disappeared. This is particularly the case when they involve investment and disinvestments mechanisms. It is a familiar point that overproduction often eventually leads to shortages.

These two examples illustrate just how complex a supposedly "simple" system is. To understand its behaviour, and to anticipate the decisions of particular actors, it is necessary to equip oneself with a systemic representation and to use system dynamics to simulate it. And what is true for a single Alpine tunnel is even more so for 2, 5, or 10 Alpine passages taking account of competition between road and rail. The simultaneous dynamics of multiply interacting sites cannot be adequately grasped without a simulator, and the interactions cannot be ignored lest the whole analysis lose all relevance.

Along the Alpine arc, it is reasonable to suppose that the actors (and how many of them are there?) each have at their disposal one or more of the action variables represented in Fig. 1. Taxes, tolls, traffic codes, investment, alternative supply: these are just some of the available tools, each naturally responding to specific objectives. It is obvious that strategic analysis, planning, and implementation of actors' policies will be all the better informed that a tool is available to simulate the overall system dynamics. Yet, in practice, such tools are hardly used at all. In fact, each actor locks itself into non-dynamic evaluations and fairly linear thinking, which inevitably precludes understanding of

the aggregate logic. To act as if what happens at the Gothard does not affect the Fréjus, the Mont Blanc or the Brenner, but only the Saint Bernard and the road, is of course a mistake. But to go further, tools are needed to shed light, *and* decision makers have to take them on board: which adds up to *two* major obstacles.

Conclusion

This article derives mainly from evidence of the growing complexity of the ways in which policies are defined and implemented when they interact with salient local issues. While reform of European Community institution is undoubtedly necessary, the same is true of national institutions in many countries. However, the institutional focus should not hide two key requirements.

First, it is imperative to develop methods of consultation and choice. This is why it appears necessary to experiment with non-exclusive, assent-based forms of expression, designed to reflect actors' preferences in a more fine-grained manner by basing votes on esteem or encouragement, voting "usefully", and rejecting the most unpopular projects or scenarios.

Secondly, evaluation must take into account, in an integrated way, trends and dynamic interactions between actors, projects, and the behaviour of the various autonomous "entities" within a system. Such evaluation requires implementation of simulation tools based on system dynamics.

In this way, the stage would be set for a more sustainable, less conflict-ridden, and more enlightened democratic system for transport, which, because of the nature of the policy issues that it involves, is an ideal arena for global thinking combined with largely decentralised implementation.

Notes

1. The Alpine Convention is a framework convention aiming at protection and sustainable development of the Alps. It was

signed on November 7 1991 in Salzburg, Austria, by Austria, France, Germany, Italy, Liechtenstein, Switzerland, and the EU.

Slovenia signed the convention on March 29 1993. An additional protocol enabled the accession of the Principality of Monaco. The

convention entered into force on March 6 1995. The “transport protocol” was signed in 2000.

2. Mr Paul Giacobbi, Corsican Assembly, session of January 25 2002: adoption of the framework document of the exceptional investment programme.

3. The preamble of the transport “services framework” states: “Set

up by the framework Law on planning and sustainable development of June 25 1999, collective services frameworks for freight and passenger transport express a new transport policy that breaks sharply with previous provisions.” Under the terms of the Law, the frameworks “determine, according to a multimodal approach, the various objectives

of passenger transport services, their modes of implementation and the criteria for selecting recommended actions, in particular with a view to ensuring long-term coherence of the networks defined for the various modes of transport and to setting their priorities with respect to operation, modernisation, adaptation and extension.”

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